Software Requirements Specification

for

Clinic Reservation System

Version 1.0

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Clinic Reservation System, a comprehensive digital solution designed to modernize and streamline appointment management within healthcare clinics. The system aims to replace inefficient manual booking processes by providing a robust platform for patients to schedule, modify, and cancel appointments, while simultaneously empowering doctors and administrative staff with efficient tools for schedule organization and patient information management.

1.2 Document Conventions

This SRS follows the IEEE Std 830-1998 recommended practice for Software Requirements Specifications. The following typographical conventions are used throughout this document:

- Bold text is used to highlight requirement attribute labels such as Description/Action, Pre-condition, Post-condition, and Output.
- Headings are hierarchically numbered (e.g., 1.1, 2.3.1) to reflect the document structure and support easy cross-referencing.
- Requirements are organized by category: User Requirements (Section 2), Functional Requirements (Section 3), and Non-Functional Requirements (Section 4).
- Each functional requirement is self-contained and carries its own implicit priority based on its order of appearance within Section 3.
- Higher-level user requirements stated in Section 2 are considered the basis from which functional requirements in Section 3 are derived. Non-functional requirements in Section 4 apply globally to the entire system unless otherwise noted.
- The abbreviation TBD (To Be Determined) is used as a placeholder where information is not yet finalized.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for the following reader types:

- Developers and Software Engineers: Should focus on Section 3 (Functional Requirements) and Section 4 (Non-Functional Requirements) for technical implementation specifications. Section 3 provides detailed pre/post-conditions and input/output definitions for each feature.
- Project Managers: Should begin with Section 1 (Introduction) and Section 2 (Overall Description) to understand scope and stakeholder needs, then consult Section 3 for deliverable breakdown and feature prioritization.
- Testers and QA Engineers: Should concentrate on the Pre-conditions, Post-conditions, and Outputs defined in each functional requirement in Section 3, as these form the basis for test case design.
- Clinical Stakeholders (Doctors and Administrative Staff): Should read Section 2.3 (User Classes and Characteristics) and Section 2 (User Requirements) which are written in accessible, non-technical language from the end-user perspective.

- Patients and End Users: Section 2.3 and the Patient Requirements subsection are most relevant, describing patient-facing capabilities of the system.

  It is recommended that all readers begin with Section 1 for a general overview, proceed to Section 2 for context and product description, and then focus on the sections most relevant to their role.

1.4 Product Scope

The Clinic Reservation System is a web and mobile-accessible software platform designed to digitize and automate appointment management for healthcare clinics. The system serves three primary user groups: patients, clinical providers (doctors), and administrative staff, each with a dedicated set of features tailored to their workflows.

Key objectives of the system include:

- Enabling patients to self-service book, reschedule, and cancel appointments at any time via web or mobile portal.
- Providing doctors with real-time schedule visibility, customizable availability blocking, and integration with Electronic Health Records (EHR).
- Equipping administrative staff with a centralized multi-doctor and multi-location dashboard for streamlined schedule management.
- Supporting automated notifications (SMS, email, push) for booking confirmations and appointment reminders.
- Managing waitlists by automatically reallocating canceled time slots to other patients.

By replacing manual scheduling processes, the Clinic Reservation System aims to reduce missed appointments, eliminate double-booking, improve staff efficiency, and ultimately enhance the quality of patient care delivered by the clinic.

1.5 References

The following documents and standards are referenced in this SRS:

1. IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications. IEEE Computer Society, 1998.
2. HIPAA (Health Insurance Portability and Accountability Act) — U.S. Department of Health & Human Services. Available at: https://www.hhs.gov/hipaa
3. GDPR (General Data Protection Regulation) — European Union, Regulation (EU) 2016/679. Available at: https://gdpr.eu
4. HL7 FHIR Standard (R4) — Fast Healthcare Interoperability Resources, used as the reference standard for EHR system integration. Available at: https://www.hl7.org/fhir
5. SRS-Template — Clinic Reservation System. Prepared by Mohamed Tarek Fathy (245180) and Ali Mohamed Hegazy (245209). ERU, March 19, 2026. (Source document for this specification.)
6. Course Materials — Instructor: Dr. Amr; Teaching Assistant: Eng. Ahmed Khattab. ERU, 2026.

2. Overall Description

2.1 Product Perspective

The Clinic Reservation System is a new, self-contained digital platform developed to replace existing manual and paper-based appointment booking processes in healthcare clinics. It is not a component of a larger pre-existing system; however, it is designed to interface with external systems, most notably Electronic Health Record (EHR) platforms, to ensure continuity and accuracy of patient data.

The system operates as a multi-user platform accessible through web browsers and mobile devices. It connects the following major subsystems and stakeholders:

- Patient Portal: A self-service interface allowing patients to register, search for doctors, book and manage appointments, and fill out medical intake forms.
- Doctor/Provider Portal: A calendar-based interface allowing doctors to manage their availability, block specific time slots, and access patient records before appointments.
- Administrative Dashboard: A centralized management panel enabling staff to oversee schedules across multiple doctors and clinic locations, handle waitlists, and manage booking confirmations.
- Notification Service: An automated engine responsible for sending SMS, email, and push notifications to patients and providers.
- EHR Integration Layer: An interface that connects the system with external Electronic Health Record databases to synchronize patient information and appointment history.

The Clinic Reservation System serves as the central hub that coordinates all appointment-related activities across these components, ensuring data consistency and operational efficiency.

2.2 Product Functions

High-level summary of the major functions the Clinic Reservation System must perform:

- User Registration and Authentication: Allow new patients and staff to create accounts and securely log in to the system.
- Appointment Booking: Enable patients to search for available doctors by specialty, location, and schedule, and book appointments online 24/7.
- Appointment Management: Allow patients to reschedule or cancel existing appointments, with the system updating availability accordingly.
- Doctor Schedule Management: Allow doctors and admins to define working hours, block specific time slots, and manage availability in real time.
- Patient Record Access: Provide authorized staff and doctors with access to patient demographic data, medical intake forms, and appointment history, linked to EHR records.
- Automated Notifications: Send booking confirmations, appointment reminders, and cancellation alerts via SMS, email, or push notification.
- Waitlist Management: Automatically offer freed-up time slots to patients on a waitlist when an appointment is canceled.
- Multi-Location Dashboard: Enable administrative staff to manage schedules for multiple doctors and branch locations from a single interface.

2.3 User Classes and Characteristics

Three primary user classes are identified for the Clinic Reservation System:

Patients
Patients are the most frequent users of the system. They interact primarily with the patient-facing portal to book, modify, and cancel appointments. Their technical expertise may range from low to moderate; therefore, the interface must be simple, intuitive, and accessible on both mobile and web platforms. Patients also need to fill out medical intake forms and upload insurance documents online prior to their visits.

Clinical Providers (Doctors)
Doctors interact with the system to manage their schedules, review patient records ahead of appointments, and block unavailable time slots for surgeries, telehealth calls, or follow-ups. They require real-time calendar synchronization and integration with EHR systems. Doctors are assumed to have moderate to high technical proficiency and interact with the system daily.

Medical Assistant / Administrative Staff
Medical assistants and administrative staff are treated as a unified user class within this system, as their responsibilities and system interactions are effectively identical in the context of clinic operations. This user class represents power users of the system. They manage schedules across multiple doctors and locations, handle waitlists, oversee booking confirmations, and assist patients who are physically present in the clinic. Medical assistants additionally support clinical workflows by preparing patient information and coordinating with doctors ahead of appointments. Both roles require access to a comprehensive, multi-functional dashboard and rely on the system's automation features to reduce manual workload. This user class holds the highest privilege level within the system.

2.4 Operating Environment

The Clinic Reservation System is intended to operate in the following environment:

- Platform: Web-based application accessible via modern desktop and mobile web browsers (Chrome, Firefox, Safari, Edge). A mobile application version is also required for iOS and Android platforms.
- Server Environment: The system will be hosted on a cloud-based infrastructure to ensure scalability, reliability, and minimal downtime.
- Database: A relational database management system (RDBMS) will be used to store user accounts, appointment records, and scheduling data.
- Operating Systems: The system must be compatible with Windows, macOS, and Linux on the server side. The client-facing portal must function on all modern operating systems through a web browser.
- External System Compatibility: The system must integrate with existing Electronic Health Record (EHR) platforms via standard healthcare interoperability protocols such as HL7 FHIR.
- Network: The system requires a stable internet connection for real-time updates, notifications, and EHR data synchronization.

2.5 Design and Implementation Constraints

The constraints will limit the options available to developers during implementation:

- Regulatory Compliance: The system must comply with HIPAA (Health Insurance Portability and Accountability Act) for patient data privacy in healthcare contexts, and GDPR (General Data Protection Regulation) for data protection of users in applicable regions.
- Security Standards: All patient and medical data must be encrypted in transit and at rest. The system must implement role-based access control (RBAC) to ensure that users only access data relevant to their role.
- EHR Integration: The system must interface with existing EHR platforms using HL7 FHIR R4 standards, which may impose constraints on data formats and API design.
- Availability: The system must maintain high availability with minimal planned or unplanned downtime, particularly during clinic operating hours.
- Cross-Platform Support: The user interface must be responsive and function correctly across all major web browsers and mobile operating systems without degradation.
- Authentication: The system must support secure user authentication mechanisms, including password hashing and optionally multi-factor authentication (MFA) for administrative accounts.

2.6 User Documentation

The user documentation components will be delivered alongside the software:

- Patient User Manual: A step-by-step guide for patients covering account registration, logging in, searching for doctors, booking and managing appointments, and filling out intake forms. Available as a downloadable PDF and accessible via an in-app help section.
- Doctor/Provider Guide: Documentation for clinical providers describing how to manage availability, block time slots, view patient records, and navigate the provider calendar. Delivered in PDF format.
- Administrator Manual: Comprehensive documentation for administrative staff covering the multi-location dashboard, schedule management, waitlist handling, and notification configuration.
- Online Help & Tooltips: Contextual in-app help tooltips and a searchable FAQ section embedded in the web and mobile interfaces.
- Quick-Start Guide: A concise one-page reference card summarizing key actions for each user class, provided in PDF format.

2.7 Assumptions and Dependencies

The assumptions and external dependencies have been identified:

- It is assumed that all end users (patients, doctors, and staff) have access to a device with a modern web browser or a smartphone capable of running the mobile application.
- It is assumed that the clinic operates with at least one existing EHR system that supports HL7 FHIR R4-compliant APIs for data integration.
- The system depends on a reliable third-party SMS and email notification service (e.g., Twilio, SendGrid) for delivering appointment reminders and confirmations. If this service is unavailable, notifications will be delayed.
- It is assumed that clinic administrators will be responsible for the initial setup of doctor profiles, working hours, and location configurations within the system.

- The system assumes stable cloud infrastructure for hosting. Any changes to the cloud provider or infrastructure setup may require reconfiguration of deployment pipelines.
- Regulatory requirements (HIPAA, GDPR) are assumed to remain as currently defined. Significant legal changes may require updates to security and privacy implementations.

3. External Interface Requirements

3.1 Hardware Interfaces

The Clinic Reservation System is a software-based solution and does not directly interface with specialized hardware components. However, the following hardware considerations apply:

- The system must function on standard consumer hardware including desktop computers, laptops, tablets, and smartphones with internet connectivity.
- The server hosting the application must meet minimum cloud infrastructure specifications for memory, CPU, and storage to support the expected concurrent user load.
- No proprietary clinic hardware (e.g., kiosk terminals) is in scope for this version of the system.

3.2 Software Interfaces

The system must interface with the following external software components:

- Electronic Health Record (EHR) System: The system must connect to clinic EHR platforms via HL7 FHIR R4 RESTful APIs to retrieve and synchronize patient medical histories, demographics, and appointment records.
- Database Management System: A relational database (e.g., PostgreSQL or MySQL) will store all application data including user profiles, appointment records, and scheduling configurations.
- Notification Services: Integration with third-party communication platforms (e.g., Twilio for SMS, SendGrid for email) for delivery of automated appointment reminders and confirmations.
- Authentication Services: The system may integrate with OAuth 2.0-based identity providers for secure, standardized user authentication.

3.3 Communications Interfaces

The communications requirements apply to the system:

- All data transmitted between the client and server must use HTTPS (TLS 1.2 or higher) to ensure encryption in transit.
- The EHR integration layer must communicate via RESTful HTTP/HTTPS APIs following HL7 FHIR R4 conventions.
- The notification system must support outbound SMTP for email and HTTP API calls to SMS gateway providers.
- Real-time schedule updates between users (e.g., when a patient books an appointment that updates a doctor's calendar) should be facilitated via WebSocket connections or equivalent real-time push mechanisms to minimize latency.

4. System Features

<This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product. You may prefer to organize this section by use case, mode of operation, user class, object class, functional hierarchy, or combinations of these, whatever makes the most logical sense for your product.>

4.1 User Registration

4.1.1 Description and Priority

This feature allows a new user (e.g., a patient) to create an account in order to access the Clinic Reservation System. Priority: High.

4.1.2 Stimulus/Response Sequences

•User navigates to the registration page on the web or mobile portal.

•User enters their full name, email address, phone number, and a secure password.

•System validates the input and checks that no active account exists for the provided email.

•System creates a new user profile, securely stores credentials, and displays a success message.

•User is redirected to the login screen.4.1.3 Functional Requirements

4.1.3 Functional Requirements

REQ-1.1	The system shall allow a new user to register by providing their full name, email address, phone number, and a secure password.
REQ-1.2	The system shall validate that no existing active account is associated with the provided email address before creating a new profile.
REQ-1.3	The system shall securely store user credentials in the database using appropriate hashing mechanisms.
REQ-1.4	Upon successful registration, the system shall display a confirmation message and redirect the user to the login screen.

4.2 Appointment Booking

4.2.1 Description and Priority

This feature enables authenticated patients to search for doctors by specialty, location, and real-time availability, and book an appointment online at any time. Priority: High.

4.2.2 Stimulus/Response Sequences

- Patient logs in and navigates to the appointment booking section.
- Patient searches for a doctor by specialty, location, or name.
- System displays available doctors with their real-time schedule.
- Patient selects a doctor, preferred date, and time slot.
- System confirms availability and locks the selected time slot.
- System displays a booking confirmation message to the patient.

4.2.3 Functional Requirements

REQ-2.1	The system shall allow patients to search for doctors based on medical specialty, location, and real-time schedule availability.
REQ-2.2	The system shall require the patient to select a doctor, desired date, and preferred time slot to complete a booking.
REQ-2.3	The selected doctor must have available time slots on the chosen date as a pre-condition for booking.
REQ-2.4	Upon successful booking, the system shall lock the selected time slot and reserve it for the patient, preventing double-booking.
REQ-2.5	The system shall display a booking confirmation message to the patient upon successful reservation.

4.3 Appointment Management

4.3.1 Description and Priority

This feature allows patients to update or cancel existing appointments. The system updates availability accordingly and manages the waitlist. Priority: High.

4.3.2 Stimulus/Response Sequences

- Patient navigates to their existing appointments in the portal.
- Patient selects an appointment and chooses to reschedule or cancel.
- For rescheduling: patient selects a new date/time; system validates and updates the booking.
- For cancellation: system cancels the appointment and makes the time slot available again.
- System notifies the patient of the updated appointment status.
- System automatically offers the freed slot to patients on the waitlist.

4.3.3 Functional Requirements

REQ-3.1	The system shall allow patients to reschedule or cancel active future appointments.
REQ-3.2	An active, future appointment must exist for the patient as a pre-condition for modification.
REQ-3.3	Upon cancellation or rescheduling, the previously reserved time slot shall be made available for other bookings.
REQ-3.4	The system shall automatically offer released time slots to other patients on the waitlist.
REQ-3.5	The system shall update the appointment status display and the master schedule accordingly.

4.4 Doctor Schedule Management

4.4.1 Description and Priority

This feature allows clinical providers and administrators to define, customize, and manage doctor availability and working hours within the system. Priority: High.

4.4.2 Stimulus/Response Sequences

- Authenticated provider or admin navigates to the calendar management dashboard.
- User defines available time slots, sets working hours, or blocks specific times (e.g., for surgeries or telehealth calls).
- System validates the input against existing bookings and schedule rules.
- System updates the doctor's schedule in real time.
- Updated calendar view is displayed to the provider.

4.4.3 Functional Requirements

REQ-4.1	The user must be authenticated with provider or administrator access privileges as a pre-condition for schedule modification.
REQ-4.2	The system shall allow providers or admins to set specific calendar dates, start/end times, and block-scheduling rules.
REQ-4.3	The system shall allow doctors to block out specific time slots for surgeries, telehealth calls, or routine follow-ups.
REQ-4.4	The doctor's calendar shall update instantly when a patient books online, preventing double-booking.

REQ-4.5	The system shall display an updated, real-time provider calendar view after any schedule change.

4.5 Patient Record Access

4.5.1 Description and Priority

This feature enables authorized clinical staff to view patient information and link scheduled appointments to patient records within the EHR system. Priority: High.

4.5.2 Stimulus/Response Sequences

- Authorized staff or doctor accesses the patient records section.
- Staff searches for a patient by name, ID, or appointment reference.
- System retrieves patient demographic data, intake forms, and appointment metadata from the EHR database.
- System displays a unified screen with the patient's full history and upcoming visits.
- Any new appointment automatically updates in both the booking system and the patient's permanent medical file.

4.5.3 Functional Requirements

REQ-5.1	A patient profile must exist and be authenticated within the system as a pre-condition for record access.
REQ-5.2	The system shall retrieve patient demographic data, intake forms, and appointment metadata from the EHR database.
REQ-5.3	The system shall display a single, organized screen showing the patient's medical history and upcoming visits.
REQ-5.4	Any new or updated appointment shall automatically synchronize with the patient's permanent medical file in the EHR system.

4.6 Automated Notifications

4.6.1 Description and Priority

This feature enables the system to automatically send booking confirmations, appointment reminders, and cancellation alerts to patients via their preferred communication channel. Priority: Medium.

4.6.2 Stimulus/Response Sequences

- An appointment reaches a 'Confirmed' state in the system.
- System retrieves the patient's contact preferences and appointment timestamp.
- Notification engine generates the appropriate message (confirmation or reminder).
- System dispatches the notification via SMS, email, or push notification.
- Notification delivery status is logged in the system.

4.6.3 Functional Requirements

REQ-6.1	An appointment must be in a Confirmed state as a pre-condition for triggering a notification.
REQ-6.2	The system shall automatically generate and send booking confirmation messages immediately after an appointment is created.
REQ-6.3	The system shall send reminder notifications to patients prior to their scheduled appointment based on the patient's contact preferences.
REQ-6.4	Notifications shall be delivered via SMS, email, or push notification depending on the patient's preferences.
REQ-6.5	Administrative staff shall be relieved of manually handling routine reminder calls and booking confirmations by the automated notification service.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall respond to user actions (e.g., searching for doctors, booking an appointment) within 3 seconds under normal operating load.
- The system shall support a minimum of 500 concurrent users without significant degradation in response time or functionality.
- Real-time calendar updates (e.g., when a booking is made) shall be reflected across all affected user views within 2 seconds.
- Automated notifications shall be dispatched within 60 seconds of the triggering event (e.g., appointment confirmation).

5.2 Safety Requirements

- The system shall ensure that no appointment data is permanently deleted without first being archived, to prevent accidental loss of medical scheduling records.
- In the event of a system failure or unexpected downtime, the system shall recover to its last consistent state and shall not create duplicate or conflicting appointment records.
- The system shall implement input validation on all user-submitted data to prevent erroneous or harmful data from corrupting scheduling or patient records.

5.3 Security Requirements

- The system shall protect all patient data and ensure privacy by complying with HIPAA (Health Insurance Portability and Accountability Act) and GDPR (General Data Protection Regulation) standards.
- All data transmitted between the client and server shall be encrypted using HTTPS with TLS 1.2 or higher.
- All stored passwords shall be hashed using a secure algorithm (e.g., bcrypt or Argon2). Plaintext passwords shall never be stored.
- The system shall implement role-based access control (RBAC) to ensure that patients, doctors, and administrative staff can only access data and functions relevant to their role.

- The system shall authenticate all users before granting access to any protected functionality or data.
- Sensitive patient health information shall only be accessible to authenticated clinical providers and authorized administrative personnel.

5.4 Software Quality Attributes

- Usability: The system shall provide a simple, intuitive, and user-friendly interface accessible on both web and mobile devices, requiring minimal training for new users.
- Reliability: The system shall maintain high availability with minimal unplanned downtime, targeting a minimum uptime of 99.5% during normal clinic operating hours.
- Scalability: The system architecture shall be designed to scale horizontally to accommodate increased numbers of users, clinics, and appointments without requiring significant rearchitecting.
- Maintainability: The system codebase shall follow established software engineering practices (e.g., modular design, documentation) to facilitate ongoing maintenance and future feature additions.
- Interoperability: The system must integrate seamlessly with external EHR platforms using HL7 FHIR R4 standards, ensuring accurate and up-to-date patient information.
- Portability: The system shall function consistently across all major modern web browsers and operating systems without platform-specific modifications.

5.5 Business Rules

- Only authenticated users with a valid account may access the appointment booking and management features of the system.
- A doctor's time slot cannot be double-booked; once reserved, it must remain locked until the appointment is canceled or completed.
- Only patients with an active account may book, reschedule, or cancel appointments.
- Only authenticated users with provider or administrator privileges may modify doctor schedules or block time slots.
- Canceled appointment slots shall be automatically offered to the next patient on the waitlist before being made publicly available.
- Patient medical records shall only be accessible to clinical providers and administrative staff with the appropriate access privileges; patients may not view other patients' records.

6. Other Requirements

The following additional requirements apply to the Clinic Reservation System:

- Database Requirements: The system shall use a relational database to ensure data integrity, support transactional operations, and maintain referential consistency between user, appointment, and schedule records.
- Internationalization: The system should be designed to support multiple languages in future versions. Text strings shall be externalized to enable localization without code changes.
- Legal Requirements: The system must adhere to all applicable national and regional healthcare data privacy regulations, including HIPAA for operations in the United States and GDPR for operations in the European Union or involving EU citizens.
- Audit Logging: The system shall maintain an audit log of all significant user actions, including appointment creation, modification, cancellation, and access to patient records, to support compliance and dispute resolution.
- Data Retention: Patient and appointment data shall be retained for a minimum period as required by applicable healthcare regulations, and a data archival mechanism shall be implemented to manage long-term storage.

Appendix : Analysis Models

The following analysis models support the requirements described in this SRS. Full diagrams are to be developed during the design phase:

- Use Case Diagram: Illustrating the interactions between Patients, Doctors, Administrative Staff, and the Clinic Reservation System for all primary use cases (booking, cancellation, schedule management, notifications).
- Entity-Relationship Diagram (ERD): Modeling the data entities and their relationships, including User, Patient, Doctor, Appointment, Schedule, Notification, and MedicalRecord.
- State-Transition Diagram: Depicting the lifecycle of an Appointment entity through states such as Pending, Confirmed, Rescheduled, Canceled, and Completed.
- Data Flow Diagram (DFD): Showing the flow of data between the Patient Portal, Doctor Dashboard, Administrative Interface, Notification Service, and EHR Integration Layer.